JAYAKODIGE LASITH CHARUKA PERERA

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SUMMARY

Final-year Bachelor of Computer Science (Artificial Intelligence) student at Swinburne University of Technology, expected to graduate in August 2026. Experienced in machine learning, computer vision, blockchain, database systems, and full-stack application development through academic and portfolio projects. Skilled in Python, Java, TensorFlow, OpenCV, SQL, Docker, Git, and data analytics, with a strong interest in building intelligent and scalable software solutions.

SKILLS

Programming: Python, Java, JavaScript, PHP, C++, HTML, CSS
Machine Learning and AI: TensorFlow, Scikit-learn, XGBoost, LSTM, GRU, CNN, Computer Vision, Grad-CAM, data preprocessing, feature engineering
Blockchain and Security: Hyperledger Fabric, Keycloak, AML monitoring, identity and access management
Databases and Analytics: MySQL, SQL, Pandas, NumPy, Matplotlib, data visualisation
Tools and Frameworks: Docker, Git, GitHub, VS Code, Jupyter Notebook, Streamlit, Flask
Professional Skills: Problem-solving, analytical thinking, teamwork, communication, time management

EDUCATION

Bachelor of Computer Science (Artificial Intelligence) - Swinburne University of Technology, Melbourne - Expected Aug 2026
Diploma in Information Technology / Computer Science - Nawaloka College - Completed

PROJECTS

Designing Australia's Central Bank Digital Currency (eAUD) - Blockchain and FinTech Capstone Project

GitHub: github.com/Dwijesh07/P29-Designing-Australia-s-Central-Bank-Digital-Currency

- Contributed to the development of a blockchain-based Central Bank Digital Currency prototype using Hyperledger Fabric.
- Implemented wallet management, transaction processing, secure authentication with Keycloak, AML monitoring, AUSTRAC compliance workflows, Docker deployment, and full-stack integration.
- Collaborated in an Agile team environment to deliver a secure fintech prototype.

AI-Powered Face Anti-Spoofing System - Computer Vision and Deep Learning Project

GitHub: github.com/jeromeliao03/COS30082-Facial-Recognition

- Designed and implemented an AI-powered anti-spoofing module for facial recognition systems.
- Trained and evaluated ResNet50 and MobileNetV2 models for liveness verification and presentation attack detection.
- Integrated Grad-CAM explainability, real-time webcam-based spoof detection, confidence scoring, and Flask/OpenCV frontend-backend integration.

Archery Score Prediction System - Machine Learning and Database Project

GitHub: github.com/lasithCharuka/Archery-score-predictor.git

- Designed and implemented a relational database system for archery score management using MySQL.
- Developed an XGBoost prediction model using historical performance data to estimate future archery scores.
- Built a Streamlit application supporting CSV uploads, real-time predictions, and user-friendly result display.
- Implemented data preprocessing, exploratory analysis, and visualisations to improve transparency of model outputs.

Stock Market Prediction Project - AI and Time-Series Forecasting Project

GitHub: github.com/lasithCharuka/Intelligent-system-cos30018/branches

- Developed stock prediction workflows using CBA.AX data from Yahoo Finance with OHLCV features and technical indicators.
- Built and compared LSTM, GRU, ARIMA, Random Forest, XGBoost, and ensemble models for financial forecasting.
- Applied scaling, sequence generation, chronological train-test splitting, and model evaluation using MAE, MSE, accuracy, precision, recall, and F1-score.
- Integrated FinBERT-based financial news sentiment analysis with SMA and RSI features.

Olympic Medal Prediction Model - Machine Learning Portfolio Project

GitHub: github.com/lasithCharuka/machine-learning.git

- Built regression-based machine learning models to predict Olympic medal counts using historical team performance datasets.
- Improved model performance through feature engineering, Ridge Regression, and evaluation tuning.
- Analysed prediction error using mean absolute error and refined the workflow through experimentation.

Weather Prediction Project - Machine Learning Portfolio Project

GitHub: github.com/lasithCharuka/machine-learning.git

- Developed a weather prediction workflow using Python, data preprocessing, and machine learning techniques.
- Applied exploratory data analysis and visualisation to understand weather patterns and key predictive features.
- Evaluated model performance and documented results to support clear interpretation of predictions.

CERTIFICATIONS

Generative AI: Working with Large Language Models | Introduction to Large Language Models | Introduction to Generative AI | Introduction to Responsible AI | Machine Learning Fundamentals

ADDITIONAL INFORMATION

- Active GitHub and portfolio website with machine learning and software development projects.
- Strong interest in Artificial Intelligence, Software Engineering, Data Analytics, Cybersecurity, and FinTech roles.
- Available for graduate, internship, junior software developer, AI/ML, data analyst, and technology graduate program opportunities.